Telescope Calculations for Real Observing

Covers Lectures 3.2–3.4 about 90 minutes

Name: _____ Date: _____ Marks: _____ / 80

Formula box

$N = F/D$ $M = F/f$ (× Barlow) $M_{\max} \approx 2D$ $M_{\min} \approx D/7$ exit pupil $= D/M$ TFOV $\approx$ AFOV$/M$
light vs eye $= (D/7)^2$ telescope vs telescope $= (D_1/D_2)^2$ Dawes $= 116/D$ arcsec $m_{\lim} \approx 2.7 + 5\log_{10} D$ (D in mm)
image size $= F \times \theta$ (rad) plate scale $= 206{,}265/F$ arcsec/mm pixel scale $= 206.265\,p\,(\mu\text{m})/F$ arcsec/px
Moon $\approx 0.5°$ Pleiades $\approx 2°$ Jupiter $\approx 45''$ Saturn with rings $\approx 40''$

Section A — Know your telescope ★ 10 marks

1. The most important specification of a telescope is its _____.
2. A _____ telescope uses a lens; a _____ telescope uses a curved mirror.
3. Newton's 1668 telescope design sends the light out of the _____ of the tube.
4. Coloured fringes around bright objects in a lens telescope are called _____ aberration.
5. A _____ mount is a simple wooden alt-az box mount for a Newtonian.
6. On an equatorial mount, the polar axis is tilted at an angle equal to your _____.
7. A _____ lens placed before the eyepiece usually doubles the magnification.
8. A **shorter** eyepiece focal length gives _____ magnification.
9. India's largest optical telescope is the _____ m telescope at _____ (near Nainital).
10. Never point a telescope at the _____ without a certified filter over the front.

Section B — The spec card ★★ 30 marks

Telescope X: Newtonian, aperture $D = 130$ mm, focal length $F = 650$ mm. Eyepieces: **25 mm** (AFOV 52°), **10 mm** (AFOV 52°), **6 mm** (AFOV 60°), and a 2× **Barlow**.

B1. [2] What is the focal ratio? Is it "fast" or "slow"?

B2. [6] Complete the table (Barlow: double the magnification; AFOV unchanged).

Eyepiece	Magnification	Exit pupil (mm)	True field (°)
25 mm			
10 mm			
6 mm			
25 mm + 2×			
10 mm + 2×			
6 mm + 2×			

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B3. [4] What are the maximum and minimum useful magnifications? Do any combinations exceed the maximum?

B4. [3] How many times more light does Telescope X collect than your dark-adapted eye?

B5. [3] What is its Dawes limit? Can it split a double star 1.5″ apart?

B6. [3] What is its limiting magnitude? Under a city sky that hides 2 magnitudes?

B7. [6] Choose the best eyepiece combination and justify in one line: (a) the whole Pleiades (2°); (b) the full Moon, as large as possible but entirely in view; (c) Saturn's rings on a steady night; (d) the faint, large Andromeda Galaxy.

B8. [3] You want ∼ 150× on Jupiter. What eyepiece focal length must you buy? Would 150× usually work under hazy Indian skies?

Section C — Compare and decide ★★ 20 marks

C1. [6] Compare Telescope X with a **70 mm f/10 refractor**: (a) the refractor's focal length; (b) with the same 10 mm eyepiece, which magnifies more — and is it the better telescope? (c) how many times more light Telescope X gathers; (d) their maximum useful magnifications.

C2. [4] A pair of 10 × 50 binoculars: (a) what do 10 and 50 mean? (b) exit pupil and light grasp vs the eye; (c) why are they ideal for the Pleiades and the Milky Way?

C3. [4] (a) What minimum aperture gives $m_{\text{lim}} = 14.4$ (Pluto)? (b) M104 is magnitude 8.0, yet often harder to see than a magnitude 11 star. Why?

C4. [3] A shop sells a 60 mm refractor labelled "450× POWER". Comment with calculations.

C5. [3] What minimum aperture splits a pair 1.5″ apart? Which instruments in this assignment can?

Section D — Astrophotography ★★★ 10 marks

A camera with **3.76 µm** pixels is placed at the prime focus of Telescope X ($F = 650$ mm).

D1. [3] How big is the Moon's image (0.52° = 0.00908 rad) on the sensor? Will it fit on a 23 × 15 mm sensor?

D2. [3] What is the pixel scale in arcseconds per pixel?

D3. [4] Jupiter is 45″ across. How many pixels wide is its image? Would the 2× Barlow help? Explain.

Section E — Practical skills (signed at the telescope workshop) 10 marks

Task	Marks	Instructor's initials
Assembled and balanced the mount safely	2	
Aligned the finder in daylight	2	
Set the latitude and pointed the polar axis north (EQ) *or* explained alt-az use (Dob)	2	
Found and focused the Moon or a planet	2	
Calculated on the spot the magnification of the eyepiece in use	2	